PROJECT ABSTRACT

RISC-V VeeR EL2 Based SoC for PWM-Controlled Peripheral Management

1. Problem Statement

Embedded systems that drive analog-equivalent loads such as motors, LEDs, heaters, and actuators require a precise, programmable, and continuously running control signal that a general-purpose processor cannot generate efficiently on its own through direct pin toggling. At the same time, such systems must remain reliable and observable: they need a mechanism to detect and recover from software or logic faults, and a simple way to display live system status without depending on an external computer connection. Implementing pulse-width control, fault recovery, and status display as separate, loosely-integrated modules increases system complexity and makes verification difficult. Therefore, there is a need for a compact, programmable System-on-Chip (SoC) that combines a processor core with dedicated, memory-mapped hardware for duty-cycle-controlled signal generation, autonomous fault detection, and local visual feedback, all coordinated through a single, well-defined architecture.

This project proposes the design and verification of a RISC-V VeeR EL2 based SoC centered on a PWM (Pulse Width Modulation) Generator for programmable duty-cycle signal control. The SoC integrates the VeeR EL2 processor with core-local instruction and data memories, standard peripherals, and three additional IPs: a PWM Generator for hardware-accurate duty-cycle waveform generation, a Watchdog Timer for autonomous system-fault detection and recovery, and a Seven Segment Display Controller for direct, human-readable status feedback. The processor manages these peripherals through a shared bus interconnect and executes an application that configures the PWM duty cycle, periodically services the watchdog, and displays the current operating status on the seven-segment display.

2. Proposed SoC Architecture

The SoC is organized around the VeeR EL2 RISC-V processor as the central control unit, connected to instruction memory (ICCM), data memory (DCCM), and peripheral IP blocks through a shared bus interconnect. The processor is responsible for configuring each peripheral through memory-mapped control and status registers, sequencing the overall flow of operations, and responding to peripheral status without needing to generate any timing-critical waveform directly in software. All additional IPs are implemented as simple memory-mapped slave peripherals on the same interconnect as the required UART, Timer, and GPIO blocks, so no additional bus-mastering logic or arbitration is required. The system follows this operational flow:

- **Duty-Cycle Control —** The processor writes a target duty-cycle value and period into the PWM Generator's control registers. The PWM hardware then free-runs independently, continuously generating the output waveform without further CPU intervention until the processor updates the registers again.
- **Fault Monitoring —** The Watchdog Timer runs a hardware down-counter that the processor must periodically reload ("kick") through a status-register write. If the processor fails to do so within the configured timeout, due to a software hang or logic fault, the Watchdog Timer autonomously asserts a system reset, allowing the SoC to recover without external intervention.
- **Status Display —** The processor writes the current PWM duty-cycle value or system status code to the Seven Segment Display Controller, which decodes the value and drives the display digits directly, giving a simple local visual readout without requiring a host computer connection.
- **Monitoring and Support —** UART provides communication between the SoC and a host computer for observing register values, watchdog events, and system status during debug, while the required GPIO and Timer support general-purpose control and independent system timing functions.

This architecture keeps the VeeR EL2 processor responsible for high-level configuration and decision-making, while the PWM Generator, Watchdog Timer, and Seven Segment Display Controller each operate autonomously in hardware once configured. Because every additional IP is a register-mapped slave peripheral with no bus-mastering or external protocol timing requirements, the overall SoC remains compact, modular, and feasible for RTL implementation and verification within a single academic term.

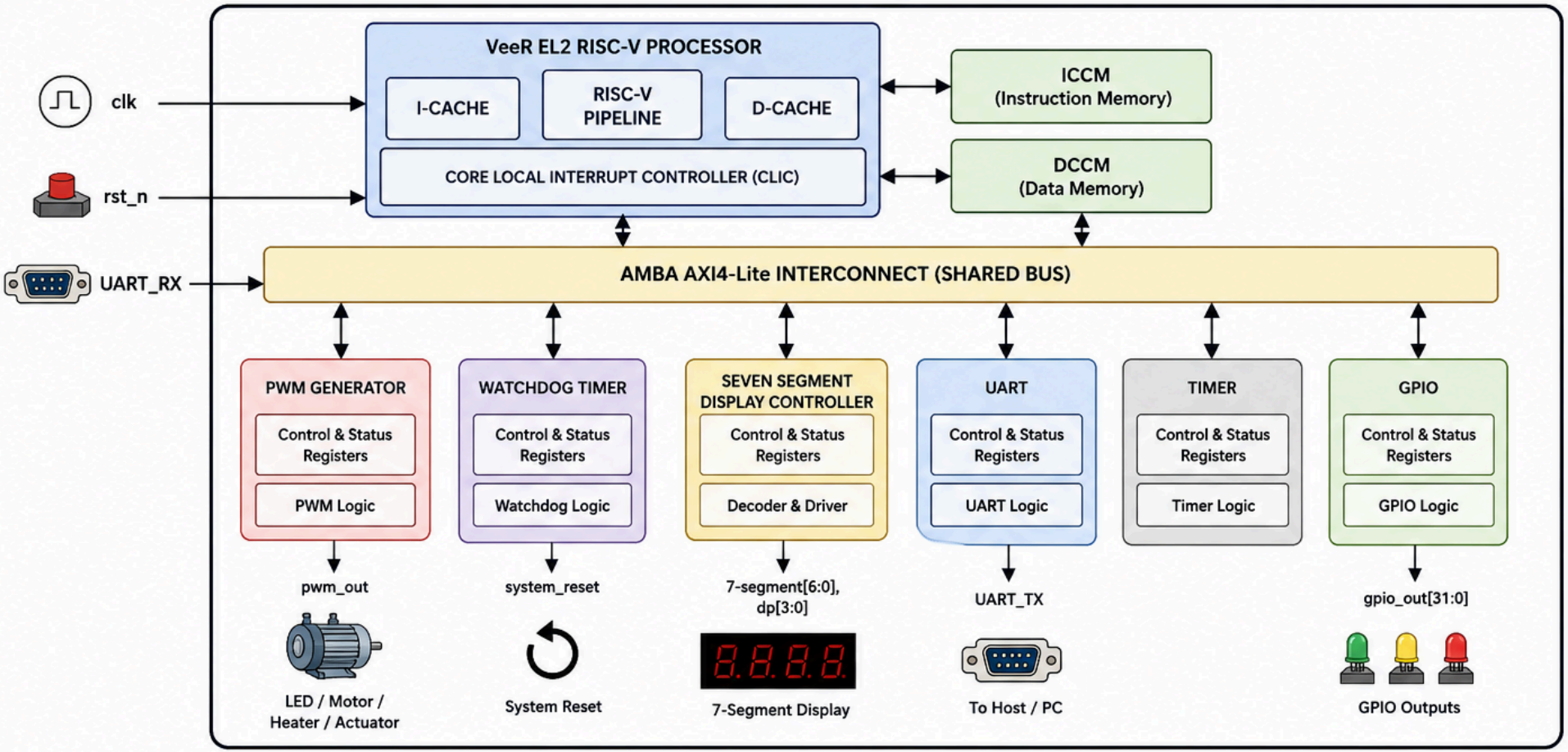
3. Selected Additional IPs

PWM Generator — The PWM Generator is the core computation IP of the SoC, producing a duty-cycle-controlled digital output waveform used to drive loads such as an LED, motor, or heater at a variable effective power level. It exposes memory-mapped period and duty-cycle registers through which the processor configures the output; internally, a free-running counter is continuously compared against the duty-cycle register to generate the output waveform in hardware, without requiring the processor to toggle the pin directly. This IP allows the SoC to demonstrate a clear, measurable hardware output while remaining simple to implement and verify.

Watchdog Timer — The Watchdog Timer protects the SoC against software hangs or unexpected control-flow faults. It is configured by the processor through a timeout register and is periodically reloaded ("kicked") by a status-register write from the running application; if this does not occur within the configured interval, the Watchdog Timer autonomously asserts a system reset. Its internal structure closely mirrors the required Timer peripheral, keeping its implementation and verification effort low while adding meaningful system-level reliability.

Seven Segment Display Controller — The Seven Segment Display Controller provides direct, human-readable visual feedback of system status without requiring a host computer connection. It accepts a value written by the processor through a memory-mapped register, decodes it into the corresponding segment pattern, and drives the display digits, using a scanning multiplexer if multiple digits are used. This gives the SoC a simple, demonstrable output for evaluating PWM duty-cycle values and system status directly on the development board.

RISC-V VeeR EL2 BASED SoC FOR PWM-CONTROLLED PERIPHERAL MANAGEMENT



INPUT SIGNALS TO THE SoC

Signal	Width	Description
clk	1	System clock input
rst_n	1	Active-low system reset
UART_RX	1	Serial receive input from external host (UART)

OUTPUT SIGNALS FROM THE SoC

Signal	Width	Description
pwm_out	1	PWM output signal to drive external load (LED/Motor/Heater)
7-segment[6:0]	7	Segment lines (a–g) for seven segment display
dp[3:0]	4	Decimal point selection for 4 digits
UART_TX	1	Serial transmit output to external host (UART)
gpio_out[31:0]	32	General purpose digital output pins
system_reset	1	System reset output from Watchdog Timer (active-high)